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Tutorial 6 Report: Arterial Stent Simulation with Balloon Inflation in ANSYS

This tutorial introduced a more complex mechanical simulation involving multiple interacting components—a balloon and an arterial stent—highlighting critical topics such as contact interactions, stepwise loading, and material selection. The main objective was to simulate balloon inflation and deflation to evaluate the mechanical performance of a stent under physiological conditions. Finite element modeling (FEM) in ANSYS was used to simulate stent deployment and recoil, allowing analysis of stress distributions, contact behavior, and the influence of material properties like Young's modulus and yield strength on residual stress and recoil.

We constructed a 3D model of an arterial stent and balloon using CAD tools embedded within ANSYS Workbench. Contact definitions were essential due to the mechanical interaction between the stent and balloon; a frictionless contact condition was applied for simplicity. The simulation was performed in two load steps—balloon inflation followed by deflation—mimicking real clinical deployment.

To ensure computational efficiency, symmetry boundary conditions were applied by modeling 1/8 of the geometry (90° arc with polar symmetry). This reduced the mesh complexity and simulation time significantly. Mesh convergence was monitored, with finer meshing near high-stress regions like the apex of the stent struts.

The deflation step proved harder to solve computationally. During inflation, the stent expands smoothly due to the internal balloon pressure. In contrast, deflation creates complex stress redistributions as the balloon contracts, resulting in non-uniform deformation and more frequent contact interactions. This complexity increased the iteration count and solution time in the deflation phase.

Using symmetry boundary conditions substantially decreased computation time without sacrificing accuracy. By reducing the geometry to 1/8 of the stent, the model maintained representative stress patterns while minimizing the number of elements, particularly in curved and high-deformation regions.

When contact interactions were suppressed and the simulation was rerun, the stent and balloon elements passed through one another, violating physical constraints. This confirmed that proper contact definition is essential in multi-body simulations, particularly in medical device modeling where precision in interaction forces is critical.

Residual stress and recoil are key indicators of long-term stent performance. Materials with higher Young's modulus resist deformation better, resulting in lower recoil but potentially higher residual stress concentrations. Conversely, lower modulus materials may deform more but distribute stress more evenly.

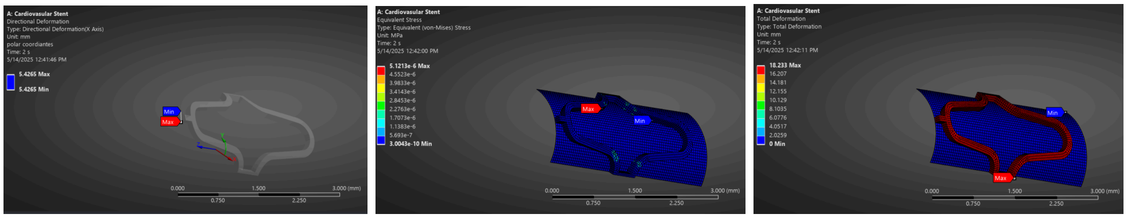

Figure 1: (From left to right) Directional Stress, Equivalent Stress, and Total Deformation of Balloon Stent Model with Stent made of modified structural steel

Replacing the generic material with accurate nitinol properties changed the mechanical response significantly. The stiffer modulus (83 GPa) led to reduced elastic deformation and better shape recovery upon balloon deflation. The yield strength (690 MPa) also increased the stent's resistance to permanent deformation, mitigating residual stress accumulation. The material was initially modeled using generic linear elastic properties, then a second simulation was run with nitinol properties based on published literature:

- Young's Modulus: 83 GPa
- Poisson's Ratio: 0.32
- Yield Strength: 690 MPa
- Density: 6450 kg/m³

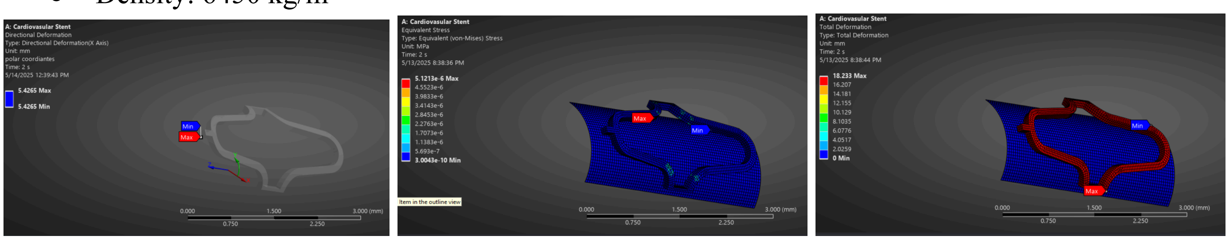

Figure 2: (From left to right) Directional Stress, Equivalent Stress, and Total Deformation of Balloon Stent Model with Stent made of nitinol

This tutorial effectively demonstrated the importance of realistic boundary conditions and material modeling in biomedical simulations. However, the computational demand of contact interactions and nonlinear material definitions could be a limiting factor for large-scale simulations. One limitation in the current model was the assumption of isotropic and linear-elastic nitinol behavior. Future iterations could incorporate superelastic or shape-memory behavior through a user-defined material model to improve realism.

Another area for refinement is simulating vessel interaction and blood flow-induced forces to more closely mimic in vivo conditions. Nonetheless, the use of a Goodman fatigue framework in literature and insights from nitinol's mechanical response provide valuable foundations for medical device performance assessment.

The arterial stent simulation reinforced key mechanical principles and introduced advanced features such as contact, multi-step loading, and material selection. The results underscore the critical role of accurate geometry, symmetry, and contact definition in realistic stent deployment modeling. Integrating material properties for nitinol allowed a deeper understanding of how elastic modulus and yield strength affect clinical outcomes like recoil and fatigue life. These skills are directly transferable to biomedical design and simulation projects involving implants or cardiovascular devices.